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Case Reports

Episcleral brachytherapy as simultaneous treatment of choroidal melanoma and neovascular membrane

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Episcleral brachytherapy is the most widely treatment for medium and small choroidal melanomas (1, 2) and it has also been proven a useful treatment for choroidal neovascularization (CNV) secondary to age-related macular degeneration (AMD) (3). The association of choroidal melanoma with CNV is rare. We presented the first case of primary association of CNV with choroidal melanoma treated simultaneously with ^{125}I episcleral brachytherapy.

Case report

The patient is a 73-year-old man with a 4 months history of visual loss and central scotoma in the right eye. The best-corrected visual acuity was 20/20 in both eyes. The right eye fundus revealed a melanotic lesion, inferotemporal to the macula, partially surrounded by extensive confluent hard exudates close to the fovea, separated from the tumor by suspected CNV (Fig. 1A). Left eye fundus examination showed neither signs of AMD nor other pathology. B scan ultrasound revealed a nodular choroidal mass, 7 mm in basal diameter and 2.6 mm of height, with choroidal excavation and orbital shadowing suggestive of choroidal melanoma. The fluorescein angiogram demonstrated fluorescence blockage by pigment and irregular choroidal vessels within the tumor bright in the early arterial phase; in addition, there were fronds of subretinal neovascularization adjacent to the tumor showed as a hyperfluorescent lesion between the posterior border of the tumor and the exudates due to occult CNV (Fig. 1B). Indocyanine green angiography showed early perifoveal hyperfluorescence establishing the presence of CNV appearing as two hot spots and a plaque lesion (Fig. 1C). Systemic work up (orbital CT scan, MR, and liver function test) was negative for metastasis. With the diagnosis of choroidal melanoma and associated CNV, episcleral brachytherapy with ^{125}I was performed as therapeutic approach for both pathologies. A Ropes plaque of 11 mm was used and dosimetric calculation was made with BEBIG plaque simulator (version 5.1.0), delivering 82.77 Gy (113.9 cGy/h) on tumor apex and 9.09 Gy (12.51 cGy/h) on the fovea. Associated CNV received from 114.99 Gy, on the edge adjacent to the tumor, to 44.02 Gy in the opposite edge (Fig. 2); treatment duration was 72.67 h. After 24 months of follow-up, tumor decreased in height up to 1.4 mm, there was marked regression of hard exudates and CNV regression (Fig. 3) and the eye has preserved vision of 20/30.

Comment

CNV has been described in association with choroidal naevi, usually as indicative of the benign nature of the choroidal lesion, or associated to choroidal melanoma after interventional treatment such as transpupillary thermotherapy. The primary association of choroidal melanoma and CNV has been reported previously only in three cases; two cases were diagnosed after enucleation (4, 5) and in the third case the diagnoses was made after membranectomy, showing in the pathologic study an unsuspected small-cell epithelioid choroidal melanoma misdiagnosed as neovascular membrane secondary to AMD. The patient received treatment with episcleral brachytherapy after developing a rhegmatogenous retinal detachment treated with pneumatic retinopexy, developing subsequently proliferative vitreoretinopathy (6). In these previous reports, the diagnosis of CNV associated with choroidal melanoma was performed after enucleation or subretinal membranectomy; however, in the present case the diagnosis of both pathologies was made before any procedure.

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